

Unit 8 Pre-Test

Transformations on the Coordinate Plane

Standards assessed: NC.8.G.2–G.4

Name: _____ Date: _____ Period: _____

SCORE **3.0** Basic Understanding

NC.8.G.2 Identify transformations and translate figures.
• *I can name translations, reflections, and rotations and translate a point.*

Match each vocabulary term to its definition, then find each image point.

– ▼ translation	A. A flip of a figure over a line.
– ▼ reflection	B. A slide; every point moves the same distance and direction.
– ▼ rotation	C. A turn of a figure about a fixed point.
– ▼ dilation	D. An enlargement or reduction by a scale factor.
– ▼ congruent	E. Same size and same shape.
– ▼ image	F. The resulting figure after a transformation.

Coordinate rules — reference

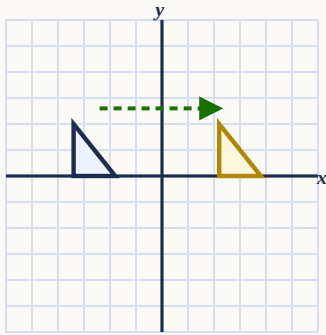
Reflect over x-axis: $(x, y) \rightarrow (x, -y)$ · Reflect over y-axis: $(x, y) \rightarrow (-x, y)$
Rotate 180°: $(x, y) \rightarrow (-x, -y)$ · Rotate 90° CCW: $(x, y) \rightarrow (-y, x)$ · Rotate 90° CW: $(x, y) \rightarrow (y, -x)$
Dilation (center origin), scale k: $(x, y) \rightarrow (kx, ky)$

1. Name it

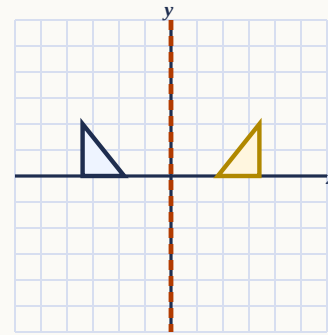
Which transformation is a slide?

2. Name it

Which transformation is a flip?



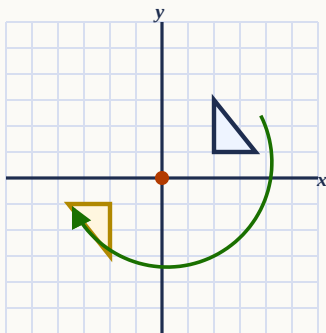
Answer: _____



Answer: _____

3. Name it

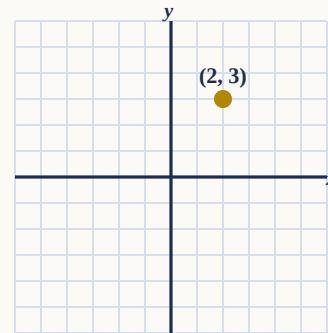
Which transformation is a turn?



Answer: _____

4. Translate

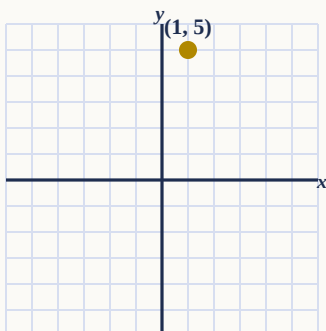
Translate $(2, 3)$ right 4 units. New point?



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5. Translate

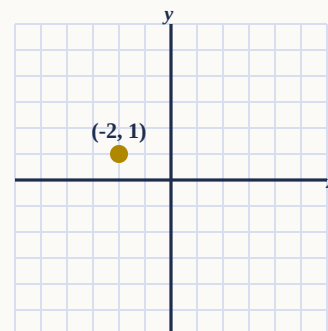
Translate $(1, 5)$ down 2 units. New point?



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6. Translate

Translate $(-2, 1)$ left 3 and up 2. New point?



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SCORE **4.0** Target Content

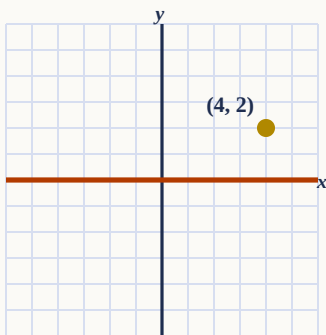
NC.8.G.3 Reflect and rotate points on the coordinate plane.

- I can find the image of a point after a reflection or rotation.

Find each image point. Write your answer as (x, y).

7. Reflect

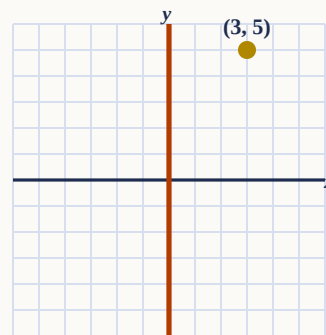
Reflect (4, 2) over the x-axis.



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8. Reflect

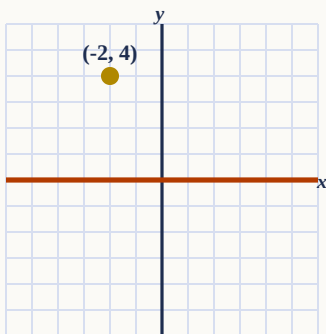
Reflect (3, 5) over the y-axis.



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9. Reflect

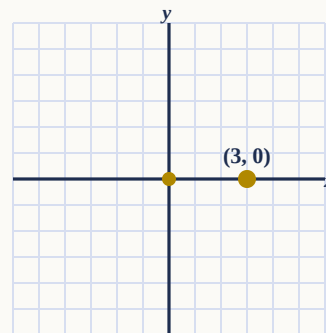
Reflect (-2, 4) over the x-axis.



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10. Rotate

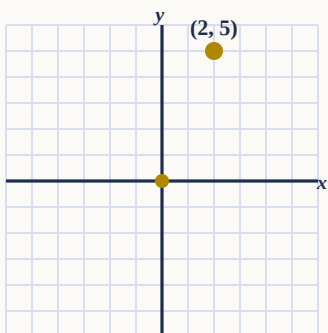
Rotate (3, 0) 90° counterclockwise about the origin.



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11. Rotate

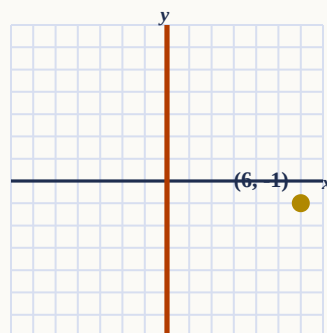
Rotate $(2, 5)$ 180° about the origin.



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12. Reflect

Reflect $(6, -1)$ over the y-axis.



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SCORE **5.0** Advanced Content

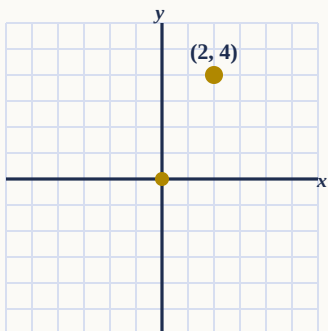
NC.8.G.4 Dilate figures and describe congruence and similarity.

- I can dilate a point by a scale factor and tell whether figures are congruent or similar.

Find each image point or answer each question.

13. Dilate

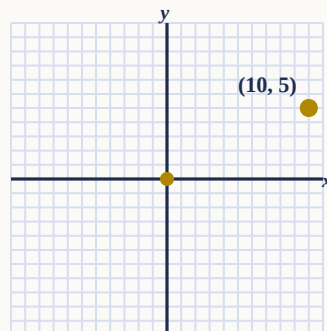
Dilate $(2, 4)$ by scale factor 3 (center origin).



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14. Dilate

Dilate $(10, 5)$ by scale factor $\frac{1}{2}$ (center origin).

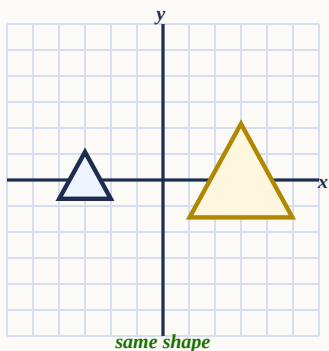


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15. Scale

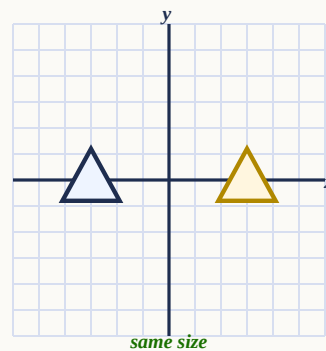
16. Congruence

A dilation with scale factor 2 makes the image ____ (larger or smaller).



Answer: _____

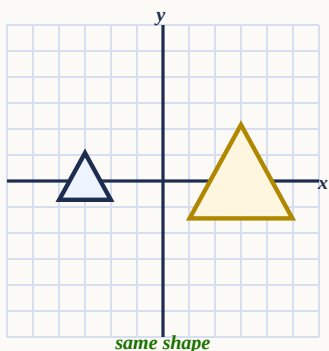
After a translation, the image is ____ to the original (congruent or similar).



Answer: _____

17. Similarity

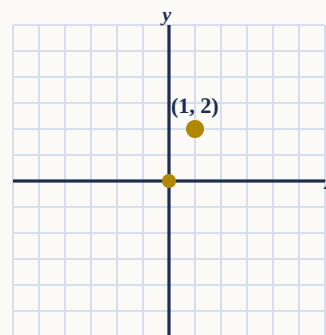
After a dilation with scale factor $\neq 1$, the image is ____ to the original.



Answer: _____

18. Rotate

Rotate $(1, 2)$ 90° clockwise about the origin.



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